MI link (Azure VM to MI) - Same region

13 May 2025 13:53

Why to use Managed Instance Link:
Enables near **real-time data replication** between SQL Server and Azure MI. The link provides hybrid flexibility and database mobility as it unlocks several scenarios, such as **scaling read-only workloads, offloading analytics and reporting to Azure, and migrating to Azure**. And, with **SQL Server 2022, the link enables online disaster recovery with fail back to SQL Server**, as well as configuring the link from SQL Managed Instance to SQL Server 2022.

How Link Works?
Private connection such as a **VPN or Azure ExpressRoute is used between an on-premises network and Azure.** If SQL Server is hosted on an Azure VM, the **internal Azure backbone** can be used between the VM and managed instance

Link: https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/managed-instance-link-feature-overview?view=azuresql#:~:text=or%20government%20clouds.-,How%20the%20link%20works,Use%20the%20link,-To%20help%20you

High Level Steps:

Step 1: Prepare Source Environment
Step 2: Prepare Target Environment
Step 3: Ensure all Pre-reqs are in place
Step 4: Steps for MI Link Creation
Step 5: Failover to MI and Failback to Azure SQL VM (conditional)

1. Create Master Key
2. Create Endpoint
3. Enable Automatic Seeding
4. Validate Connection for 5022,11000-11999 & 1433 are able to communicate with each other (MI-SQL Server)
   Test-NetConnection targetsqldb01.bfe2fe49b482.database.windows.net -port 5022
5. Create MI Link

Pre-req:

1. An active Azure subscription. If you don't have one, create a free account.
2. Supported version of SQL Server with required service update installed.
   Link: https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/managed-instance-link-feature-overview?view=azuresql#:~:text=Initial%20primary%20version,2016%20aren%27t%20supported.
3. Azure SQL Managed Instance. Get started if you don't have it.
4. SQL Server Management Studio v19.2 or later.
5. A properly prepared environment.
6. Managed Instance will use V-net local Endpoint.

Consider the following:
Link: https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/managed-instance-link-configure-how-to-ssms?view=azuresql#:~:text=Consider%20the%20following,SQL%20Server%202022.

1. The link feature supports **one database per link**.
2. Collation between SQL Server and SQL Managed Instance should be the same.
3. Error 1475 on your initial SQL Server primary indicates that you need to start a new backup chain by creating a full back up without the COPY ONLY option.
4. While you can establish a link from SQL Server 2022 to a SQL managed instance configured with the **Always-up-to-date update policy,** after failover to SQL Managed Instance, you will no longer be able to replicate data or fail back to SQL Server 2022.

Permission on Source and Target:

1. Source Azure VM : sysadmin permission on instance
2. MI: SQL Managed Instance contributor or customer rule:

Step 1: Create VM on azure and Managed Instance (Same Region)
Step 2: Backup restore few databases on Az VM and changed the recovery mode to full.
Step 3: Post changing the recovery model take full backup (Azure VM)
Step 3: Install Az.SQL module: --Optional (to use Powershell over GUI)
Install-Module -Name Az.Sql

Note: Configure link with SSMS - Azure SQL Managed Instance | Microsoft Learn

Use the link feature to replicate databases from your initial primary to your secondary replica. For SQL Server 2022, the initial primary can be either SQL Server or Azure SQL Managed Instance. For SQL Server 2019 and earlier versions, the initial primary must be SQL Server.

If you plan to use your secondary managed instance for only disaster recovery, you can save on licensing costs by activating the AzHB.

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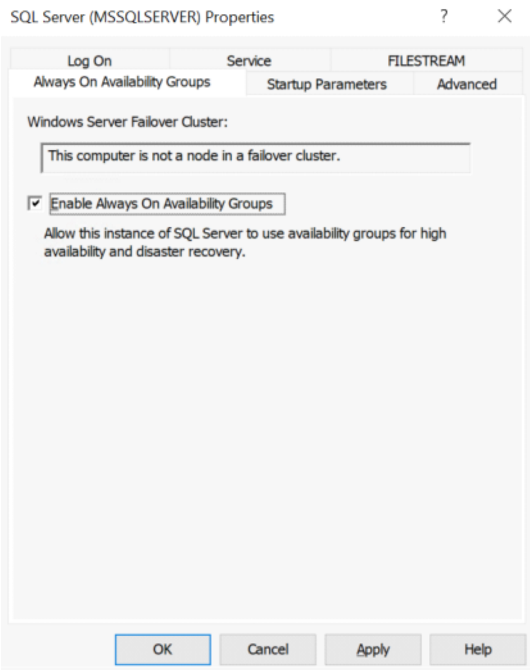
Prepare database on Source (Azure VM)
Create a backup of your database (In case of Azure VM).
Since Azure SQL Managed Instance takes backups automatically. Skip this step if SQL Managed Instance is your initial primary.

Server readiness | Availability group readiness

Action	Result
SQL Server: Version check	Ready
SQL Server: Always On Availability ...	Always On Availability Groups not enabled
SQL Server: Sysadmin role membe...	Ready
SQL Server: Recommended trace fl...	Recommended trace flag 1800 not enabled
SQL Server: Recommended trace fl...	Recommended trace flag 9567 not enabled

How to resolve server readiness issues :
SQL Server: Always ON Availability Groups Enabled



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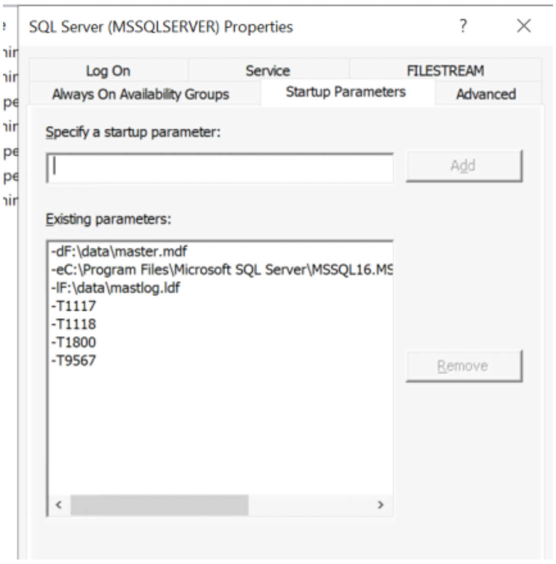
Trace Flag 1800 and 9567 Enabled



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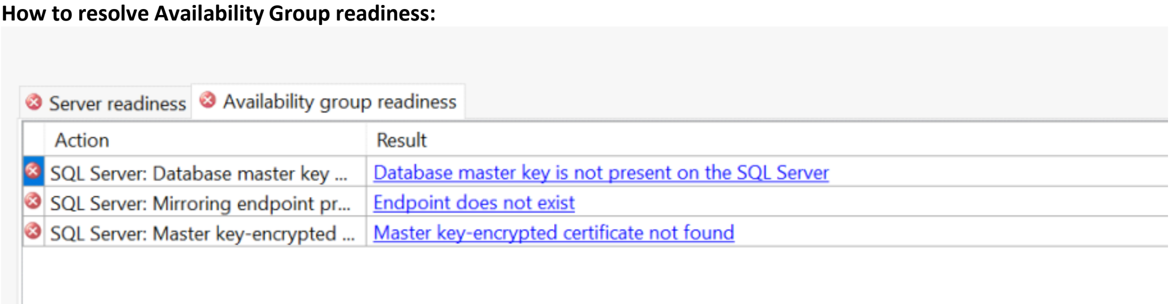
How to resolve Availability Group readiness:



Note: Master Key we need to create on Source. For, Endpoint and Master key-encryption certificate it will auto-create at the time of MI Link configuration.

Why to use DB Master Key for MI link?
Used for **encryption and secure communication** in SQL Server environments.
To encrypt credentials or **use certificates for secure communication with an MI**, a **DMK must be created** in the master database or in the user database depending on context.

Run select to find the certificate Key and Certificate Key --

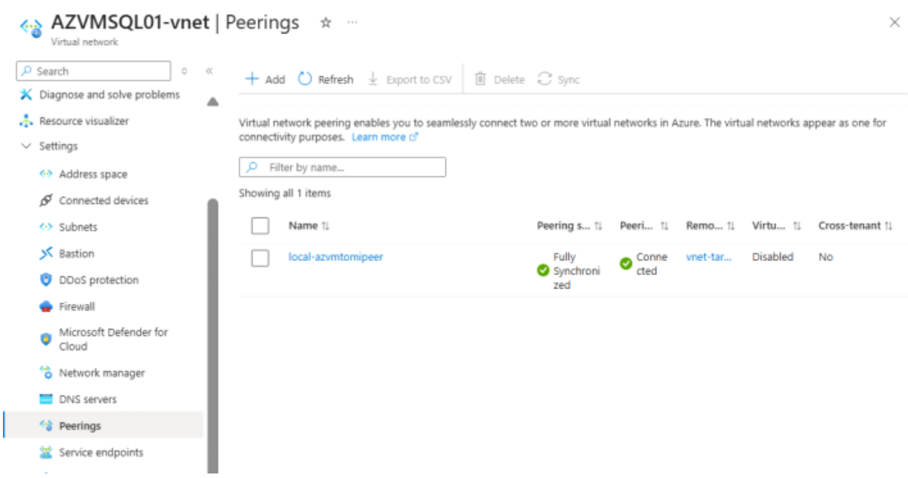
USE master;
GO
SELECT * FROM sys.symmetric_keys WHERE name LIKE '%DatabaseMasterKey%';

--Create Master Key on both Source Servers (Pre-requisites for MI Link Creation) --
--Note: Ensure to maintain the password
USE master;
GO
CREATE MASTER KEY ENCRYPTION BY PASSWORD = '';
Note: Testing@1234567890

Test Connection to MI Link: It failed showing Vnet peering is not in place.

Vnet Peering (Same Region so need to enable on first 2)
Description:
Virtual network peering enables you to seamlessly connect two or more virtual networks in Azure. This will allow resources in either virtual network to directly connect and communicate with resources in the peered virtual network.

Note: Diff region/ Env : Need to create VPN gateway



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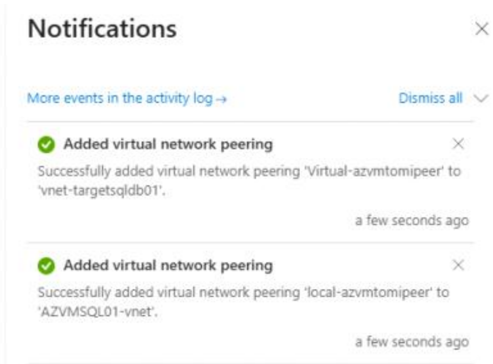

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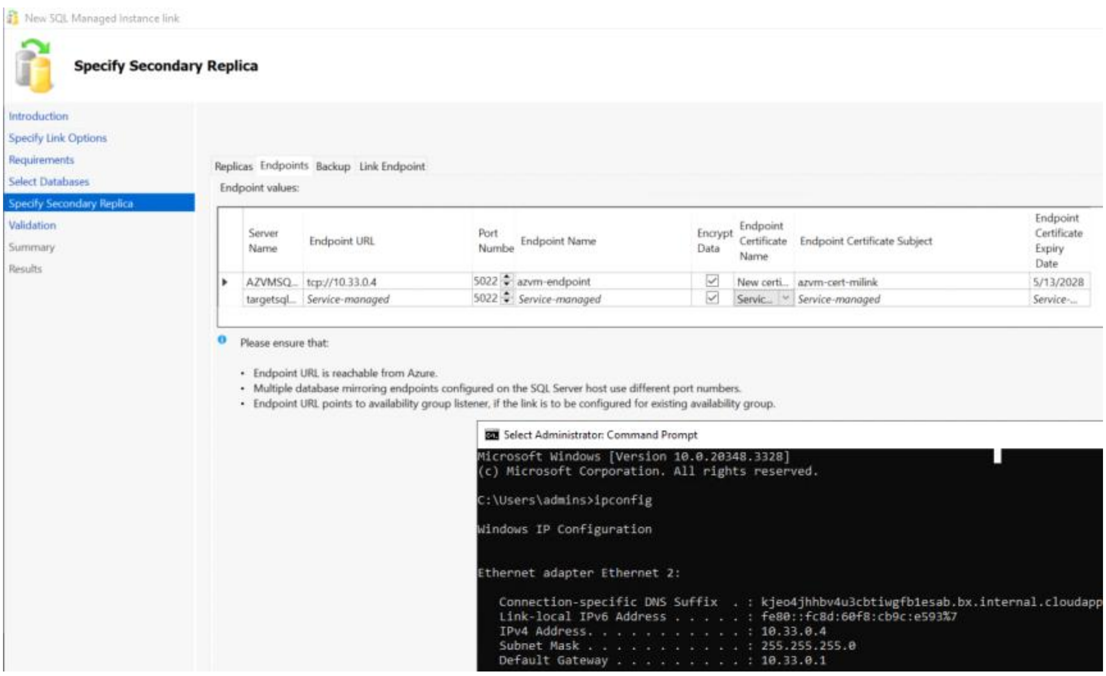
Configuring MI link:



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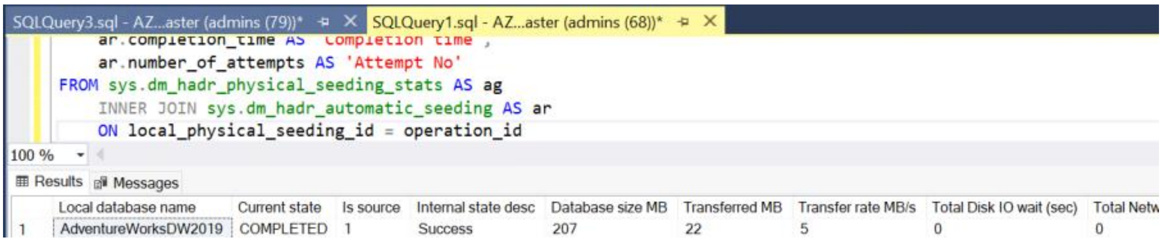
-- OPTIONAL
-- You can use the following query to monitor initial database seeding progress from SQL Server

```
SELECT
   ag.local_database_name AS 'Local database name',
   ar.current_state AS 'Current state',
```

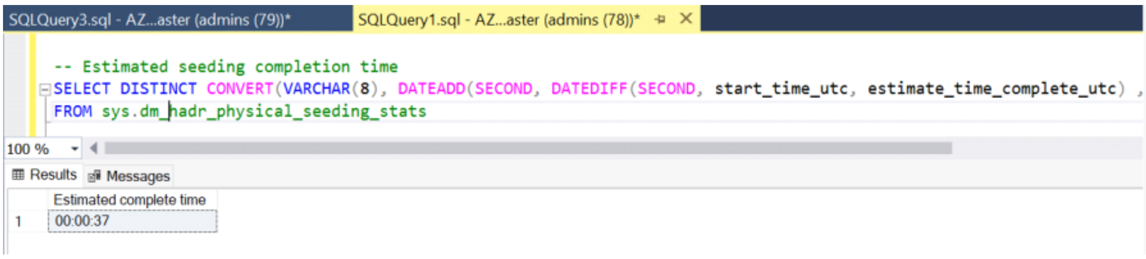
```
    ar.is_source AS 'Is source',
    ag.internal_state_desc AS 'Internal state desc',
    ag.database_size_bytes / 1024 / 1024 AS 'Database size MB',
    ag.transferred_size_bytes / 1024 / 1024 AS 'Transferred MB',
    ag.transfer_rate_bytes_per_second / 1024 / 1024 AS 'Transfer rate MB/s',
    ag.total_disk_io_wait_time_ms / 1000 AS 'Total Disk IO wait (sec)',
    ag.total_network_wait_time_ms / 1000 AS 'Total Network wait (sec)',
    ag.is_compression_enabled AS 'Compression',
    ag.start_time_utc AS 'Start time UTC',
    ag.estimate_time_complete_utc as 'Estimated time complete UTC',
    ar.completion_time AS 'Completion time',
    ar.number_of_attempts AS 'Attempt No'
FROM sys.dm_hadr_physical_seeding_stats AS ag
    INNER JOIN sys.dm_hadr_automatic_seeding AS ar
    ON local_physical_seeding_id = operation_id
```



	Local database name	Current state	Is source	Internal state desc	Database size MB	Transferred MB	Transfer rate MB/s	Total Disk IO wait (sec)	Total Netw
1	AdventureWorksDW2019	COMPLETED	1	Success	207	22	5	0	0

-- Estimated seeding completion time

```
SELECT DISTINCT CONVERT(VARCHAR(8), DATEADD(SECOND, DATEDIFF(SECOND, start_time_utc, estimate_time_complete_utc) ,0), 108) as
'Estimated complete time'
FROM sys.dm_hadr_physical_seeding_stats
```



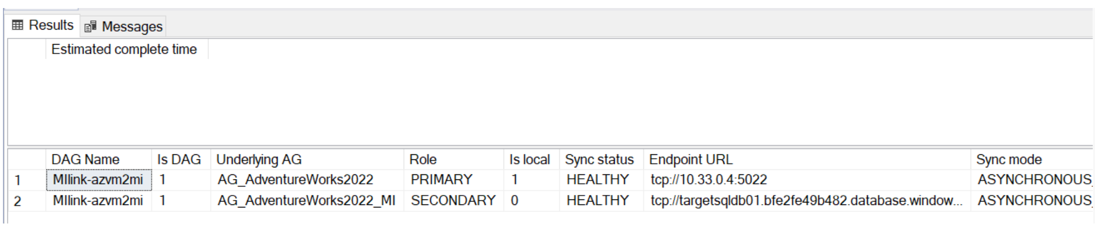
	Estimated complete time
1	00:00:37

--- OPTIONAL

-- You can use the following query to monitor the sync status of the link during its lifetime

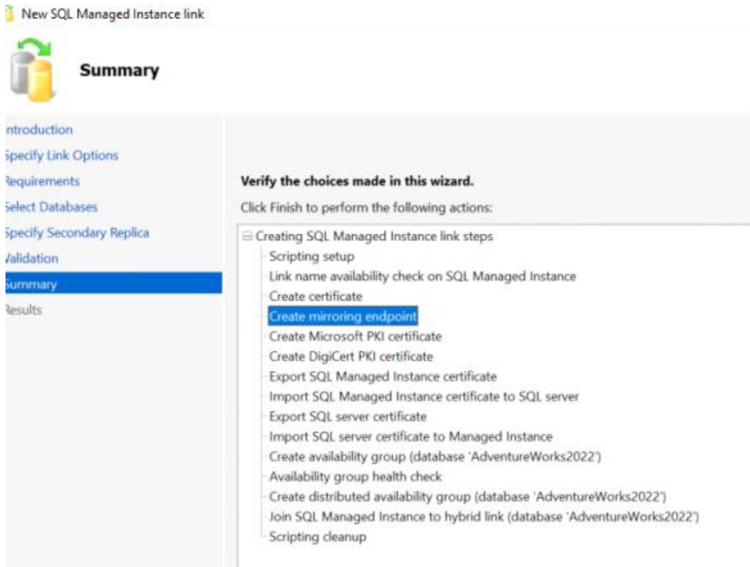
```
SELECT
    ag.[name] AS [DAG Name], ag.is_distributed AS [Is DAG], ar.replica_server_name AS [Underlying AG], ars.role_desc AS [Role],
    ars.is_local AS [Is local], ars.synchronization_health_desc AS [Sync status], ar.endpoint_url as [Endpoint URL],
    ar.availability_mode_desc AS [Sync mode], ar.failover_mode_desc AS [Failover mode], ar.seeding_mode_desc AS [Seeding mode],
    ar.primary_role_allow_connections_desc AS [Primary allow connections], ar.secondary_role_allow_connections_desc AS [Secondary allow
connections]
FROM  sys.availability_groups AS ag
INNER JOIN sys.availability_replicas AS ar
    ON  ag.group_id = ar.group_id
INNER JOIN sys.dm_hadr_availability_replica_states AS ars
    ON  ar.replica_id = ars.replica_id
WHERE ag.is_distributed = 1

GO
```

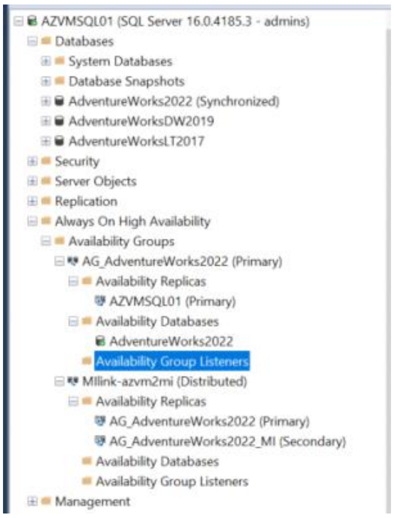


	DAG Name	Is DAG	Underlying AG	Role	Is local	Sync status	Endpoint URL	Sync mode
1	MIlink-azvm2mi	1	AG_AdventureWorks2022	PRIMARY	1	HEALTHY	tcp://10.33.0.4:5022	ASYNCHRONOUS
2	MIlink-azvm2mi	1	AG_AdventureWorks2022_MI	SECONDARY	0	HEALTHY	tcp://targetsqldb01.bfe2fe49b482.database.window...	ASYNCHRONOUS

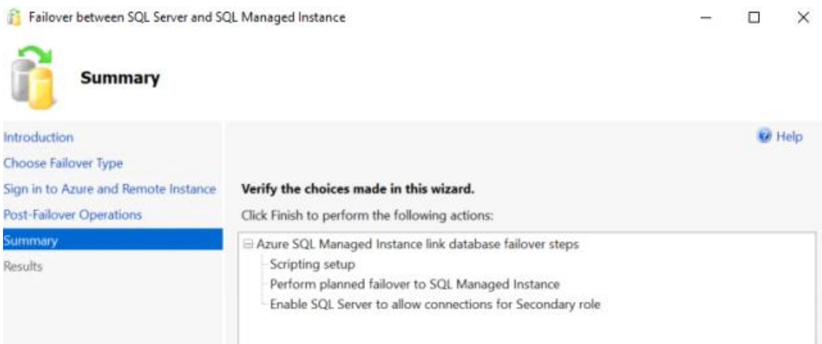
Execution for MILink Status (Azure VM to SQL Managed Instance)



Validate Always On High Availability



Failover to MI



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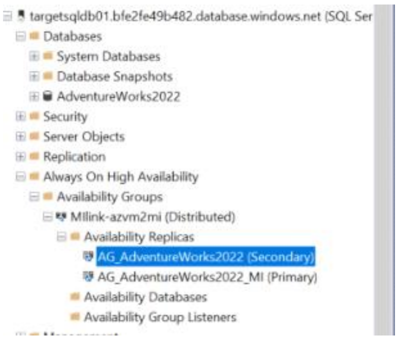
Managed Instance status post failover:



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